

63V 3A CHARGER SPECIFICATION

102003001204-V1.0

1、系统框图 System Diagram

2、参数表 Parameter List

NO.	参数 PARAMETER	规格 SPEC	
1	输入特性 INPUT PROPERTY		
1.1	空载最大功耗 No load maximum power consumption	<1W	
1.2	输入电压范围 Input voltage range	AC90-138Vac	
1.3	输入频率范围 Input frequency range	47~63HZ	
1.4	效率 Efficiency	>85%	
1.5	浪涌电流 Inrush current	Inrush Current<50A @138Vac	
2	输出特性 OUTPUT PROPERTY		
2.1	最高输出电压 Rated output voltage	64.3±0.5V	
2.2	额定待机电压 Rated Standby voltage	35V±0.5V	
2.3	额定输出电流 Rated output current	3±0.3A	
2.4	充满截止的电压/电流 Full cut-off voltage/current	63V±0.20V， 500±200mA	
2.5	滞后恢复充电电压 Hysteresis recovery charging voltage	最高单节电压小于 4.0±0.05V	
2.6	充电器匹配电池包的规格 Charger matching battery pack specification	63V battery pack	
2.7	涓流 Trickle-charge 电池电压在涓流电压范围内，不开继电器以涓流电流进行充电，在涓流时间内如还未达到涓流结束电压，充电器停止充电并指示异常	涓流电压 1.0-2.0±0.1NV 涓流电流 100mA 以下 涓流时间 30min±5%	
2.8	预充 Pre-charge 电池电压在预充电电压范围内，以预充电电流充电，在预充时间内如还未达到预充结束电压，充电器停止充电并指示异常 Charging within the pre-charge current when the pack voltage in the range of the pre-charge voltage, after pre-charge time, if battery’s voltage is still less than the end pre-charge voltage, the charger stop charging the battery pack and indicate abnormal.	63V battery pack	
		预充电电压 Pre-charge voltage	1.0---2.5±0.1*NV
		预充电电流 Pre-charge current	500±200mA
		预充时间 Pre-charge timer	30min±5%
2.9	反向电流（无 AC 电源时插入电池包） Reverse current(Battery in charger with no AC power applied)	<10uA	
2.10	正常充电温度 Normal charge temperature	根据电池包配置:（18650: -2±3℃---50±3℃ 21700: -2±3℃---57±3℃）	
2.11	滞后恢复温度	根据电池包配置:（18650: 0±3℃---45±3℃ 21700: 0±3℃---50±3℃）	
2.12	输出纹波和噪声 Output ripple and noise	≤220mVP-P	
3			
3.1	充电器接入电源，电池包不插入充电器 Charger connects to power supply. Battery pack does not insert the charger	红色 LED 常亮,绿色 LED 熄灭 Red LED on,Green LED off	

3.2	充电器给电池包充电中 On charging	红色 LED 熄灭, 绿色 LED 1Hz 频率闪烁 Red LED off Green LED flashes at a frequency of 1Hz
3.3	检测到电池包电压异常 Abnormal voltage of battery	红色 LED 1Hz 频率闪烁, 绿色 LED 熄灭 Red LED flashes at a frequency of 1Hz, Green LED off;
3.4	电池包温度异常 (温度保护模式) Abnormal temperature of battery (temperature protection mode)	红 LED 和绿色 LED 1HZ 交替闪烁 Red LED and green LED flash alternately at a frequency of 1 Hz;
3.5	电池包充饱, 充电器关闭充电 Battery is fully charger, charger stops charging	红色 LED 熄灭, 绿色 LED 常亮 Red LED off, Green LED on
4	保护特性 PROTECTION PROPERTY	
4.1	低压保护 Battery pack abnormal protection	当电池包电压低于 $1.0 \pm 0.2 * N V$ 时, 关断输出, 报错 When the battery pack voltage is below $2.0 \pm 0.2 * N V$, cut off the output
4.2	过流保护(1P) Over current protect(5S1P)	当充电电流超过 $3.5 \pm 0.3 A$, 关断输出, 报错 When the charging current is more than $3.5 \pm 0.3 A$, cut off the output
4.3	过压保护 Over voltage protect	当电池电压超过 $4.2 \pm 0.02 * N V$, 关断输出, 显示充满 When battery' s voltage is more than $4.2 \pm 0.02 * N V$, cut off the output
4.4	短路保护 Short circuit protect	当输出端短路时, 充电器不能损坏 When short circuit occurs in output end, charger shall not be broken
4.5	温度过高/过低保护 Over/Under temperature protect	当温度超过 $57 \pm 3^{\circ} C$ 或低于 $-2 \pm 3^{\circ} C$ (21700), 超过 $50 \pm 3^{\circ} C$ 或低于 $-2 \pm 3^{\circ} C$ (18650) 时, 停止充电 Stop charging when the temp. is either more than $57 \pm 3^{\circ} C$ or less than $-2 \pm 3^{\circ} C$ (21700), more than $50 \pm 3^{\circ} C$ or less than $-2 \pm 3^{\circ} C$ (18650);
4.6	通信异常保护 Communication failure protect	当电池包插入时通信异常, 切断输出 When battery plugged in and communication abnormal, cut off the output
4.7	定时关断保护 Timing cut off protect	充电超时后自动切断: 容量/电流*60min*1.25 倍+30min
5	端口 PORT	
5.1	Port 1: B+ Port 2: Charge positive C+ Port 3: COM Port 4: Charge negative B-	
6	环境和其他 ENVIROMENT AND OTHERS	
6.1	温度 Temperature	Operating: 5 --- $50^{\circ} C$ Storage: $-25^{\circ} C$ --- $+60^{\circ} C$
6.2	湿度 Humidity	Operating: 20% --- 95% Storage: 20% --- 90%
6.3	绝缘阻值 Insulation resistance	DC500V, 1min 1.insulation among housings, output end and input end) $> 20 M \Omega$ 2.insulation between output end and housing $> 2 M \Omega$
6.4	耐电压 Dielectric withstand voltage	额定电压 $\leq 150 V$ Rated voltage $\leq 150 V$ 安规测试标准/safety test standard : 1250V/5mA/60S

	电压实施在输入端子和输出端/外壳螺丝 The voltage is applied between AC input terminals and DC output terminals/shell/screw		生产测试标准/Production test standard : 1250V/5mA/3S
		150V< 标准电压 $\leq$ 250V 150V<Rated voltage $\leq$ 250V	安规测试标准/safety test standard : 3750V/5mA/60S 生产测试标准/Production test standard : 3750V/5mA/3S
6.5	保护解释 Protection interpretation	当充电器保护功能动作时, 不得有输出电压和电流, 即完全关闭 When the charger protection function is closed, there shall be no output voltage and current, namely totally closed.	
6.6	安全,认证 Safety,Certification	CE\UL1012\PSE	

3、功能说明 Function Specification

3.1 LED

3.1.1 指示灯: 红绿指示灯各一个

Indicating led: a red led, a green led.

3.2 充电逻辑 Charging principle

3.2.1 当充电器插入市电时, 红灯指示灯常亮, 绿灯熄灭; 当电池包插入时, 通过电压判断电池包是否插入

When the charger is applied to the power supplier, red led is on and green led is off; Check whether battery pack is plugged in through voltage.

3.2.2 当电池包电压低于 1*N_V 时, 关断输出, 红灯闪烁

When the battery pack voltage is below 1*N_V, cut off the output, the charger will go into abnormal state with red led blinking.

3.2.3 当电池电压在 1-2.5V 每节时, 预充电流充电, 此时红灯熄灭, 绿灯闪烁, 如果 30 分钟无法充到 2.5V 每节以上, 进入异常状态: 绿灯熄灭, 红灯闪烁

When battery's voltage is between 1*N-2.5*N V, a current of Pre charge is used for charging with red led off and green led blinking; If battery's voltage can't reach above 2.5*N V after 30min charging, the charger will go into abnormal state with green led off and red led blinking.

3.2.4 当电池电压大于 2.5V 每节时, 进入到大电流 CC 充电, 此时红灯熄灭, 绿灯闪烁

When battery's voltage is more than 2.5*N V, a current of CC is used for charging with red led off and green led blinking.

3.2.5 当电测到电池电压充到设定电压 4.20*N V, 逐渐减少充电电流, 此时红灯熄灭, 绿灯闪烁; 当电流减少到关断电流时, 指示灯指示充满, 完全切断充电

When battery's voltage reaches 4.20*N V, charging current will be reduced gradually with red lamp off and green lamp blinking; Until the charging current is reduced to end current, cut off the output.

3.2.6 当进入到温度异常状态时, 温度恢复后可自行充电; 当进入到电流或电压异常状态, 需重新插拔电池后才能再次开启检测

When the overcharge data received from battery, the charger will cut off output and indicate fully charged.

3.2.7 温度从异常值恢复后, 充电器将自动重新开始充电。如果电流或电压或通信是不正常的, 电池组必须重新被插入, 并再次运行

After the temperature recover from abnormal value, the charger will automatically start charging again. If the current or voltage or communication is abnormal, the battery pack has to be plugged out and in to operate again.

4、版本修改及放行记录 Version revision and release records

NO.	版本编号 Version No.	修改内容 Modify Content	修改者 Modified By	日期 Date
1	1.0	1. 初次发行 Initial release		

